

Evaluating Order Hypotheses Within Proximity Matrices

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We discuss the task of evaluating conjectures of order within proximity matrices in connection with an inappropriate inference procedure proposed by Wakefield and Doughtie and subsequently adopted by other authors. This strategy is criticized and a different method based on randomization is proposed. An example from Holland's theory of personality types is used throughout this article for illustration. Finally, we develop the testing of two competing theoretical conjectures against a given proximity matrix as a natural extension of evaluating one such hypothesis.

Most of the standard multivariate procedures intended for the analysis of correlation or covariance matrices are used in an exploratory manner, even though a researcher's basic intent might be to verify a precisely specified substantive conjecture. Typically, theory evaluation relies on loose heuristic comparisons between whatever empirical results are obtained from a strategy of unrestricted analysis and whatever patterns should be present in the given data if the theory is appropriate. There are obviously notable exceptions to this statement when confirmatory factor analysis and some of the extensions to the analysis of covariance structures are considered, but in general, this summary is a reasonable assessment of how multivariate methods are commonly applied in the social and behavioral sciences.

One illustration used throughout this article is from vocational psychology, namely, Holland's (1966, 1973) theory of careers based on six personality types—realistic (R), investigative (I), artistic (A), social (S), enterprising (E), and conventional (C). Given an inventory that provides scores on six scales corresponding to these six personality types (e.g., the Strong-Campbell Vocational Interest Inventory or the Holland Vocational Preference Inventory), the intercorrelations between types for a sample of subjects should reflect the interpoint distances within the circular structure presented in Figure 1. For instance, in an earlier empirical test of the Holland structure, Cole and Hanson (1971) applied a type of principal-component analysis to several such six-by-six correlation matrices, plotted the six types based on two components, and concluded that the resulting plots were consistent with the representation specified in Figure 1. Although other authors may rely on different methods of spatial representation (e.g., see Coleman, 1970; Rounds, Davison, & Dawis, 1979), the general approach is still the same.

Although simple visual inspection might be considered the norm, it is not surprising that the strong structural hypotheses

derived from Holland's model (or similar substantive theories) have also motivated attempts to provide more direct tests. Wakefield and Doughtie (1973), in particular, suggested a hypothesis-testing strategy for conjectures of order within a correlation matrix. That approach allegedly allows the evaluation of whether such precise structures are manifest in the data. It is this latter method, which has since been adopted without expressed reservations by others (e.g., Bobele, Alston, Wakefield, & Doughtie, 1979; Gati, 1982; Rounds et al., 1979; Tuck & Keeling, 1980) that we question. More pointedly, the Wakefield-Doughtie procedure is incorrect, and even though we agree with the need for exact techniques of evaluation when blessed with precise theory, the inferential procedure used must have statistical legitimacy.

Given this brief introduction, the present article is organized as follows: First, we introduce the task of testing order hypotheses within a correlation matrix by using a broad framework that allows the evaluation of order hypotheses in any "proximity" matrix (defined as any matrix containing numerical measures of the interrelations between the objects from the set defining its rows and columns). We then critically review the Wakefield-Doughtie (1973) procedure and propose a different strategy that could lead to a justifiable test of the Holland conjecture as well as similar types of precisely defined structures. We end by discussing a method that allows the testing of two competing theoretical conjectures against a single proximity matrix.

Specifying an Order Hypothesis Within a Proximity Matrix

Instead of concentrating on covariance or correlation matrices, a discussion of testing structural (order) hypotheses can be specified without loss of generality by using a generic proximity measure defined on pairs of objects from some set $S = \{O_1, O_2, \dots, O_n\}$. Our usage of the term *proximity* originates from terminology in scaling (Coombs, 1964; Kruskal & Wish, 1978; Shepard, 1962) and merely refers to some arbitrary numerical measure of correspondence between distinct pairs of objects from S . For convenience, the proximity between O_i and O_j will be denoted by p_{ij} , and the resulting $n \times n$ proximity matrix, containing p_{ij} as the entry in row i and column j , by $P = \{p_{ij}\}$.

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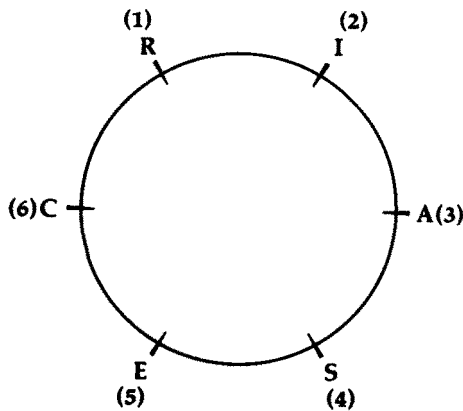


Figure 1. A representation of the circular model for Holland's (1966, 1973) six personality types. (The integers are assigned to the types according to the text.)

Proximity is assumed to be symmetric and nonnegative (i.e., $p_{ij} = p_{ji}$ and $p_{ij} \geq 0$ for $1 \leq i, j \leq n$) and is keyed as a dissimilarity measure so that larger positive values of p_{ij} suggest greater degrees of dissimilarity between O_i and O_j . Thus, if r_{ij} refers to the product-moment correlation between O_i and O_j , $1 - r_{ij}$ would have the form of a symmetric dissimilarity measure; in fact, this measure is the one used in the example given below. For our purposes, the main diagonal of P , that is, p_{ii} for $1 \leq i \leq n$, will be irrelevant and can be set to zero arbitrarily.

A given structural hypothesis will be specified by an order relation on pairs of entries from P . More explicitly, the given structural hypothesis is assumed to be formalized by either a three- or a four-argument function, $T_H(\cdot, \cdot, \cdot)$ or $F_H(\cdot, \cdot, \cdot, \cdot)$, defined on $S \times S \times S$ or $S \times S \times S \times S$, respectively (where the subscript, H , mnemonically refers to the hypothesized order relation and T and F , to *three* and *four*), as follows:

$$T_H(O_i, O_j, O_k) = \begin{cases} 1 & \text{if } p_{ij} \text{ is conjectured to be} \\ & \text{smaller than } p_{ik}, i \neq j \text{ and } i \neq k; \\ 0 & \text{otherwise,} \end{cases}$$

$$F_H(O_i, O_j, O_k, O_l) = \begin{cases} 1 & \text{if } p_{ij} \text{ is conjectured to be smaller} \\ & \text{than } p_{kl}, i \neq j \text{ and } k \neq l; \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

In an illustration using the conjecture of circularity in Figure 1, Wakefield and Doughtie (1973) argued that if such a structure is appropriate, then as a necessary implication, the correlations for the six adjacent object pairs (RI, IA, AS, SE, EC, and CR) should be greater than the correlations for the six alternate pairs (RA, AE, ER, IS, SC, and CI). In turn, the correlations between alternate pairs should be greater than those for the three opposite pairs (IE, RS, and CA). Thus, if we assign the objects O_1, O_2, O_3, O_4, O_5 , and O_6 to the six personality types R, I, A, S, E, and C, respectively, the Wakefield-Doughtie conjecture leads immediately to a four-argument function, $F_H(\cdot, \cdot, \cdot, \cdot) = F_H$. For instance, $F_H(O_1, O_2, O_1, O_3) = 1$ because

(O_1, O_2) corresponds to the adjacent pair (R, I) and (O_1, O_3) corresponds to the alternate pair (R, A). Similarly, the function F_H can be defined for every quadruple of objects from S ; that is, $F_H(O_i, O_j, O_k, O_l) = 1$ if the pair (O_i, O_j) is adjacent and (O_k, O_l) is alternate or opposite or if (O_i, O_j) is alternate and (O_k, O_l) is opposite.

In another illustration of how conjectures may be formalized for the Holland theory, Gati (1979) concentrated on the set of order relations defined by a three-argument function, $T_H(\cdot, \cdot, \cdot) = T_H$, which forms a subset of those considered by Wakefield and Doughtie (1973). In particular, $T_H(O_i, O_j, O_k) = F_H(O_i, O_j, O_i, O_k)$, implying that $T_H(O_i, O_j, O_k) = 1$ if and only if (O_i, O_j) is adjacent (alternate) and (O_i, O_k) is alternate or opposite (opposite). Here, all comparisons are mediated by one object, O_i , common to the two pairs (O_i, O_j) and (O_i, O_k) . Thus, hypothesized conjectures are made only within the rows of P .

In addition to the hypothesized functions T_H and F_H , analogous definitions can be constructed for three- and four-argument functions defined from P , $T_D(\cdot, \cdot, \cdot)$ and $F_D(\cdot, \cdot, \cdot, \cdot)$, where D mnemonically stands for proximities *data*, as follows:

$$T_D(O_i, O_j, O_k) = \begin{cases} 1 & \text{if } p_{ij} \text{ is smaller than } p_{ik} \\ & i \neq j \text{ and } i \neq k; \\ 0 & \text{otherwise.} \end{cases}$$

$$F_D(O_i, O_j, O_k, O_l) = \begin{cases} 1 & \text{if } p_{ij} \text{ is smaller than } p_{kl}, \\ & i \neq j \text{ and } k \neq l; \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

Thus, a simple count of the number of conjectures verified by the data could be given by the cross-products:

$$\sum_{i,j,k} T_H(O_i, O_j, O_k) T_D(O_i, O_j, O_k)$$

and

$$(1/4) \sum_{i,j,k,l} F_H(O_i, O_j, O_k, O_l) F_D(O_i, O_j, O_k, O_l).$$

The factor of 1/4 eliminates redundancy in this latter term. The four quadruples (O_i, O_j, O_k, O_l) , (O_j, O_i, O_k, O_l) , (O_i, O_j, O_l, O_k) , and (O_j, O_i, O_l, O_k) must produce exactly the same values for F_H and F_D . Similarly, the four quadruples (O_k, O_l, O_i, O_j) , (O_k, O_l, O_j, O_i) , (O_l, O_k, O_i, O_j) , and (O_l, O_k, O_j, O_i) are equivalent. In effect, the factor of 1/4 chooses one representative from each of these two groups to define the total measure, for example, (O_i, O_j, O_k, O_l) and (O_k, O_l, O_i, O_j) . No such redundancy exists when using the three-argument functions because each pair of object pairs with a common object is already considered only twice; that is, the triple (O_i, O_j, O_k) corresponds to (O_i, O_j, O_i, O_k) , which contributes in this form and as (O_i, O_k, O_i, O_j) .

As a simple illustration of these two descriptive expressions, Table 1 presents one minus the intercorrelations for 2,433 women between the six personality types when using the Vocational Preference Inventory, taken from the *Handbook for the ACT Career Planning Program* (American College Testing Program, 1968), as reported by Rounds et al. (1979). As can be assessed from Table 1, there are 11 violations of the 72 order

Table 1

One Minus the Intercorrelations for 2,433 Women Between the Six Personality Types When Using the Vocational Preference Inventory (as Reported by Rounds, Davison, & Dawis, 1979)

Type	R	I	A	S	E	C
R	.00	.46	.64	.70	.57	.71
I	.46	.00	.56	.68	.74	.92
A	.64	.56	.00	.60	.59	.98
S	.70	.68	.60	.00	.54	.80
E	.57	.74	.59	.54	.00	.48
C	.71	.92	.98	.80	.48	.00

Note. R = realistic; I = investigative; A = artistic; S = social; E = enterprising; C = conventional.

conjectures characterized by F_H and 6 violations of the 48 order conjectures characterized by T_H . Specifically, for T_H , the following pairs are in violation:

$$\{(1, 6), (1, 3)\}, \{(1, 6), (1, 4)\}, \{(1, 6), (1, 5)\}, \\ \{(2, 6), (2, 5)\}, \{(3, 4), (3, 5)\}, \text{ and } \{(4, 6), (4, 1)\}.$$

When considering F_H , the following five additional pairs contribute:

$$\{(1, 6), (2, 4)\}, \{(1, 6), (3, 5)\}, \{(3, 4), (1, 5)\}, \\ \{(4, 6), (1, 4)\}, \text{ and } \{(4, 6), (2, 5)\}.$$

The obvious task at this point is to evaluate the relative magnitude of these descriptive measures.

As a summary thus far, we start with some given algebraic structure presumed to underlie and generate the proximity matrix P . If this hypothesis is appropriate, then certain order relations should be present; that is, the given structure leads to certain necessary conditions that are specified as constraints on possible orderings among the entries in P ; the latter are characterized formally by three- or four-argument functions, T_H or F_H . The number of agreements between the conjectured orders and the data in the proximity matrix P may then be counted through cross-product statistics comparing T_H and F_H with the corresponding functions T_D and F_D obtained from P . The next task is to assess formally the relative magnitude of these latter summary measures.

Alternative Structural Hypotheses

Although we do not provide explicit numerical illustrations here, it should be apparent how structures other than the circular form of Figure 1 might lead to a definite set of order hypotheses on the entries in P that could be similarly formalized. In fact, any precisely defined geometric or algebraic representation is a candidate. For example, suppose a linear ordering of the objects along a continuum is conjectured, and objects O_i and O_j should be placed at locations x_i and x_j ($x_i < x_j$). Then, p_{ij} is conjectured to be smaller than p_{kl} whenever $x_k < x_i < x_j \leq x_l$, leading directly to a four-place function, $F_H(O_i, O_j, O_k, O_l)$; or for the corresponding three-place function, $T_H(O_i, O_j, O_k)$, p_{ij} is conjectured to be smaller than p_{ik} whenever $x_i < x_j < x_k$. If

the structure assumed is not linear but, instead, consists of a set of subsets of S , denoted by $S_1, \dots, S_m$ (where $1 \leq m < n$), $F_H(O_i, O_j, O_k, O_l)$ could be generated by conjecturing that p_{ij} will be smaller than p_{kl} whenever O_i and O_j belong to the same subset S_h for some h but that O_k and O_l do not belong to a common subset; $T_H(O_i, O_j, O_k)$ could be obtained by conjecturing that p_{ij} is less than p_{ik} whenever both O_i and O_j belong to some S_h for some h but O_k does not.

Testing an Order Hypothesis Within a Proximity Matrix

Wakefield-Doughtie inference strategy. In attempting to assess the magnitude of the number of violations of the conjecture specified by F_H (and by implication by T_H as well), Wakefield and Doughtie (1973) proposed a simple binomial test. Their null assumption can be stated as follows: Given a pair of object pairs for which a conjecture of order is made, the probability that the conjecture holds is $1/2$, and the probability that the conjecture fails to hold is also $1/2$. Then, if N denotes the number of object pairs for which order conjectures are made, the probability of obtaining exactly r violations (and, conversely, $N - r$ agreements) is $\binom{N}{r} (.5)^N$, and thus, the p value (one-tailed) is defined as

$$\sum_{r=0}^v \binom{N}{r} (.5)^N,$$

where v is the observed number of violations. For example, in our two illustrations of the last section, there are 72 conjectures of order for F_H and 11 violations observed, with 36 expected; for T_H , there are 48 conjectures of order and 6 violations observed, with 24 expected. Thus, p values less than .001 would be obtained by using the normal approximation to the binomial for z values of -5.89 and -5.20 , respectively.

One minor gesture acknowledging redundancy was Wakefield and Doughtie's not considering conjectures of order to be made explicitly between adjacent and opposite pairs. The latter are implicitly generated by transitivity once the relations between adjacent and alternate pairs and between alternate and opposite pairs are specified. Thus, Wakefield and Doughtie would consider 54 conjectures of order for F_H and 10 violations observed, with 27 expected; for T_H , there would be 48 conjectures of order and 5 violations observed, with 24 expected, again giving p values less than .001 by using the normal approximation to the binomial for z values of -4.63 and -4.33 , respectively. Unfortunately, merely eliminating redundant pairs definable by transitivity does not obviate our objections to those authors' procedure.

The major flaw in the Wakefield-Doughtie (1973) strategy arises from the structure that must exist among the conjectured order relations, even though the "redundant" relations between adjacent and opposite pairs might not be considered. For example, in considering this reduced set of 54 conjectures for F_H , these authors' null hypothesis assumes that all possible patterns of successes and failures of the predictions are equally likely. This latter statement is unjustifiable, and one simple counterexample suffices as a demonstration. Suppose that in the 54 conjectures of order, there are two given violations— $p_{RA} > p_{RS}$ and $p_{RI} > p_{RA}$ —and one nonviolation— $p_{SR} > p_{SI}$. Then, by neces-

sity, $p_{IR} > p_{IS}$. In other words, there must be a violation for the pair IR and IS; thus, patterns of successes and failures that have these two violations and the nonviolation must have a probability of zero unless $p_{IR} > p_{IS}$. This same counterexample would be appropriate for T_H , given the manner in which it was constructed.

The basic criticism of Wakefield and Doughtie's (1973) procedure concerns their assumption of independent binomial realizations to evaluate the given set of order conjectures. Unfortunately, the realizations of such order relations are usually not independent because the violation or nonviolation of some order relations necessarily implies the violation or nonviolation of others. If an adequate hypothesis-testing strategy is to be defined, the structure implicit in using pairs has to be taken into account explicitly. (The Wakefield-Doughtie procedure is in contrast to Kendall's, 1970, legitimate use of this same binomial null assumption to evaluate the number of circular triads in paired comparison matrices. Here, the interest is specifically in rejecting this null model for a particular subject by comparing the observed number of circular triads with the expected number under the binomial model when $p = .5$. If our interest had focused on comparing a given order with the subject's data, however, the binomial null model would not have been appropriate for the same reasons the Wakefield-Doughtie strategy can be criticized.)

Inference strategy based on random relabeling. When used to test hypothesized order relations within a proximity matrix P , the Wakefield-Doughtie (1973) strategy will undoubtedly be flawed because of the equally likely requirement for the predicted pattern of successes and failures. Thus, because this equally likely requirement is the basis for the inference strategy, all applications of the Wakefield-Doughtie method are suspect even when applied in contexts other than evaluating the Holland model; for example, see the applications to the Illinois Test of Psycholinguistic Ability discussed by Doughtie, Wakefield, Sampson, and Alston (1974).

We present an approach to the evaluation of conjectured orders within a proximity matrix not suffering from these difficulties. Specifically, consider the cross-product measure providing the number of agreements of the conjectured orders as given by F_H :

$$(1/4) \sum_{i,j,k,l} F_H(O_i, O_j, O_k, O_l) F_D(O_i, O_j, O_k, O_l).$$

This observed number of agreements will be evaluated with respect to the distribution of such agreements if the n objects in S had actually been labeled at random.

Formally, consider the cross-product

$$(1/4) \sum_{i,j,k,l} F_H(O_i, O_j, O_k, O_l) F_D(O_{\rho(i)}, O_{\rho(j)}, O_{\rho(k)}, O_{\rho(l)}),$$

where $\rho(\cdot)$ is a one-to-one permutation of the first n integers. This latter cross product gives the number of agreements to the conjectured orders if the particular 0-1 order relation initially hypothesized for the quadruple (O_i, O_j, O_k, O_l) is now evaluated against the data in P for the quadruple $(O_{\rho(i)}, O_{\rho(j)}, O_{\rho(k)}, O_{\rho(l)})$. When tabled over all $n!$ possible permutations, considered equally likely, a distribution over the number of agree-

Table 2

Cumulative Distributions of the Two Cross-Product Statistics Based on the Data of Table 1

Cumulative frequency	Three arguments	Four arguments
1	12	19
5	12	19
10	12	19
20	13	19
50	15	22
100	17	25
200	20	30
300	23	33
400	24	37
500	27	42
600	30	46
700	36	52
708	36	52
709	42	61
720	42	61

ments is obtained under the null hypothesis that the observed permutation is selected at random. If the observed number of agreements is "too large" with respect to this latter distribution, the null hypothesis of random relabeling is rejected. The p value associated with the observed number of agreements is the proportion of the $n!$ permutations that give agreement counts as large or larger.

Similarly, the observed cross-product statistic based on T_H and T_D may be evaluated. Again, the null hypothesis of interest is one of a random relabeling of the objects on which the order conjecture is defined. Thus, given the manner in which the distribution is generated, the structural integrity of the order conjecture is preserved for each realization within the null distribution, in contrast to the Wakefield-Doughtie (1973) method.

To illustrate this strategy, Table 2 presents the cumulative distribution of the cross-product statistic based on $6! = 720$ permutations for both the three- and four-argument alternatives. Using 6 and 11 violations (or 42 and 61 agreements), respectively, both p values are $12/720 = 1/60 = .02$, suggesting that the number of conjectured order relations that are violated is significantly less than would be obtained under a random relabeling of the objects (or, equivalently, that the number of agreements is too large). In fact, the numbers of agreements for both the three- and four-argument alternatives are at their maxima over all 720 permutations. (With respect to this particular circular structure, there are really only $5!/2 = 60$ "distinct" relabelings because all cyclic permutations and their complete reversals are equivalent. Because a distribution that is equally likely over all 720 permutations induces an equally likely distribution over the 60 distinct relabelings, we could have restricted ourselves to this latter set.)

An interesting connection exists between the random relabeling strategy presented here and the Wakefield-Doughtie (1973) technique whenever there are no tied proximity values in P . Even though the distributions obtained under each procedure are very different, the expected number of agreements is the same. For example, since there are no tied proximities in Table 1, the three- and four-argument cross-product expressions have expectations, respectively, of (see Hubert, 1978):

$$\begin{aligned}
& \{1/[n(n-1)(n-2)]\} \\
& \quad \times \sum_{i,j,k} T_D(O_i, O_j, O_k) \sum_{i,j,k} T_H(O_i, O_j, O_k) \\
& = \{1/(6 \cdot 5 \cdot 4)\} \{(1/2)6 \cdot 5 \cdot 4\} \\
& \quad \times \{\text{number of three-argument order conjectures}\} \\
& = \{1/2\} \{48\} = 24
\end{aligned}$$

and

$$\begin{aligned}
& \{1/[n(n-1)(n-2)]\} \\
& \quad \times \sum_{i,j,k} F_D(O_i, O_j, O_i, O_k) \sum_{i,j,k} F_H(O_i, O_j, O_i, O_k) \\
& + \{1/n(n-1)(n-2)(n-3)\} \\
& \quad \times \sum_{\substack{i,j,k,l \\ (\text{distinct})}} F_D(O_i, O_j, O_k, O_l) \sum_{\substack{i,j,k,l \\ (\text{distinct})}} F_H(O_i, O_j, O_k, O_l) \\
& = \{1/2\} \{\text{number of three-argument order conjectures} \\
& \quad + \text{number of four-argument order conjectures not} \\
& \quad \text{subsumed under three-argument order conjectures}\} \\
& = \{1/2\} \{48 + 24\} = 36.
\end{aligned}$$

These latter values are the same as those obtained under the inappropriate binomial assumption. Although this equivalence of expectations may seem anomalous, the reason for it is fairly simple. A random relabeling conjecture actually compares each given order conjecture made for a particular object triple or quadruple with all possible relabelings of the objects in that triple or quadruple. Because exactly half (assuming no ties in P) of the relabelings will satisfy any specific order conjecture when compared with the relevant proximities, the overall expectation must be the same as for the binomial. As noted, however, the distributions themselves (e.g., variances) would be very different.

Extensions and Discussion

Descriptive Measures of Correspondence Between Conjectured Orders and Observed Orders

In addition to a significance test based on the magnitude of the number of violations of the order conjecture, it is also possible to develop normalized descriptive statistics that indicate the degree to which the conjecture is satisfied. Moreover, these descriptive measures possess convenient probabilistic interpretations with respect to the observed data and, thus, confer an operational meaning on the observed level of the measure.

Suppose in addition to counting the number of agreements and violations of the order conjecture, we also obtain the number of instances in which a decision could not be reached, that is, when an order conjecture was made but the relevant proximities in P were tied. If we refer to the number of violations as D , the number of nonviolations as A , and the number of null decisions as T (for disagreements, agreements, and ties, respectively), we propose a descriptive measure of the form

$$\frac{A - D}{A + D + T}. \quad (3)$$

This latter quantity varies between the bounds of ± 1 and has a nice probabilistic interpretation with respect to P . Specifically, $(A - D)/(A + D + T)$ is the difference between two conditional probabilities, $A/(A + D + T)$ and $D/(A + D + T)$. If we assume that a quadruple or triple for which a specific order conjecture has been made (there are $A + D + T$ such conjectures) is picked at random, the first term, $A/(A + D + T)$, is the probability that the order conjecture is satisfied; $D/(A + D + T)$ is the probability that the order conjecture is not satisfied. Again, using our example, for F_H we would obtain

$$\begin{aligned}
(61 - 11)/72 &= (61/72) - (11/72) = .85 - .15 \\
&= .69 \text{ (up to rounding),}
\end{aligned}$$

and for T_H we would obtain

$$(42 - 6)/48 = (42/48) - (6/48) = .88 - .13 = .75.$$

Thus, in addition to rejecting the null hypothesis of a random relabeling, substantial descriptive differences exist in the conditional probabilities of obtaining an agreement versus a disagreement with an order conjecture (i.e., differences of .69 and .75).

To represent the descriptive measure in Equation 3 formally through a cross-product statistic, suppose the three- and four-argument functions are modified as follows:

$$T_H(O_i, O_j, O_k) = \begin{cases} 1 & \text{if } p_{ij} \text{ is conjectured to be} \\ & \text{smaller than } p_{ik}, i \neq j \text{ and } i \neq k; \\ -1 & \text{if } p_{ij} \text{ is conjectured to be} \\ & \text{larger than } p_{ik}, i \neq j \text{ and } i \neq k; \\ 0 & \text{otherwise,} \end{cases}$$

$$F_H(O_i, O_j, O_k, O_l) = \begin{cases} 1 & \text{if } p_{ij} \text{ is conjectured to be} \\ & \text{smaller than } p_{kl}, i \neq j \text{ and } k \neq l; \\ -1 & \text{if } p_{ij} \text{ is conjectured to be} \\ & \text{larger than } p_{kl}, i \neq j \text{ and } k \neq l; \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

with T_D and F_D defined analogously with respect to the proximity matrix P . Then, the normalized measures having the form in Equation 3 are given by

$$\begin{aligned}
& \sum_{i,j,k} T_H(O_i, O_j, O_k) T_D(O_i, O_j, O_k) / \sum_{i,j,k} |T_H(O_i, O_j, O_k)|, \\
& \sum_{i,j,k,l} F_H(O_i, O_j, O_k, O_l) F_D(O_i, O_j, O_k, O_l) / \\
& \quad \sum_{i,j,k,l} |F_H(O_i, O_j, O_k, O_l)|, \quad (5)
\end{aligned}$$

where $|x|$ denotes the absolute value of x . Under the random-labeling hypothesis and following the derivations given by Hubert (1978), normalized measures of the form given in Equation 5 have expectations of zero.

In summary, measures of the form given in Equation 3 are bounded between ± 1 , have expectations of zero under the ran-

Table 3
Cumulative Distributions of the Two Normalized Descriptive
Statistics Based on the Data of Table 1—
Holland's (1966, 1973) Model

Cumulative frequency	Three arguments	Four arguments
1	-.50	-.47
5	-.50	-.47
10	-.50	-.47
20	-.46	-.47
50	-.38	-.39
100	-.29	-.31
200	-.17	-.17
300	-.04	-.08
400	.00	.03
500	.13	.17
600	.25	.28
700	.50	.44
708	.50	.44
709	.75	.69
720	.75	.69

dom-labeling hypothesis, and may be understood descriptively by considering the random selection of order conjectures from the given data. In fact, instead of considering the raw measure A by itself, as we did earlier, we may wish to evaluate the normalized measure directly. As can be seen from the distributions under the random-labeling conjecture given in Table 3 for our example, the two normalized measures and the raw number of agreements are exact linear transformations of each other; that is, the two normalized measures could be obtained as $(A/36) - 1$ for the four-argument functions and as $(A/24) - 1$ for the three-argument functions. This result stems from the absence of tied proximity values in P , and thus, T is 0 for all $6! = 720$ permutations. If there are tied proximity values in P and, depending on how T_H and F_H are defined, an exact linear relation may not exist between the two distributions because T may itself vary over the $n!$ permutations. But one would still expect a very good, although approximate, linear relation to obtain.

Computational Issues in the Random Relabeling Strategy

As can be seen from the example just given, it is possible to develop an exact distribution for the number of agreements to the conjectured orders when the data matrix P is reasonably small (e.g., $n \leq 8$). For situations requiring larger data matrices, the computational burden imposed by complete enumeration is very large, and alternative methods must be sought. One of the more appealing methods is a Monte Carlo significance-testing approach in which a random sample of the $n!$ permutations is used to obtain a significance level for the observed number of agreements (or similar descriptive measure). Here, N permutations are chosen at random (with replacement), and if M of the associated measures are as extreme or more so when compared with the observed agreement measure, a p value (one-tailed) of $(M + 1)/(N + 1)$ is reported. (The 1 appears both in the numerator and the denominator because the observed agreement measure is assumed to be another random draw under the null as-

sumption.) In short, other than the imposition of the initial sampling process, the evaluation of an observed measure is the same as when an exact distribution is available. A more complete discussion of Monte Carlo significance testing can be found in Edgington's (1980) and Cliff and Ord's (1981) books.

Comparing Two Competing Order Conjectures

One of the more noteworthy extensions of the strategy we have proposed to assess the reasonableness of a single theory would be the comparison of two competing theories, referred to as H_1 and H_2 , and evaluation of which might be closer to the data. If $F_{H_1}(O_i, O_j, O_k, O_l)$ and $F_{H_2}(O_i, O_j, O_k, O_l)$ or $T_{H_1}(O_i, O_j, O_k)$ and $T_{H_2}(O_i, O_j, O_k)$ formalize these two competing conjectures as ± 1 functions analogous to Equation 4, consider the cross products

$$(1/4) \sum_{i,j,k,l} F_{H_1:H_2}(O_i, O_j, O_k, O_l) T_D(O_i, O_j, O_k, O_l)$$

and

$$\sum_{i,j,k} T_{H_1:H_2}(O_i, O_j, O_k) T_D(O_i, O_j, O_k),$$

where

$$F_{H_1:H_2}(O_i, O_j, O_k, O_l)$$

$$= \begin{cases} 1 & \text{if } F_{H_1}(O_i, O_j, O_k, O_l) = 1 \\ & \text{and } F_{H_2}(O_i, O_j, O_k, O_l) = -1, \\ -1 & \text{if } F_{H_1}(O_i, O_j, O_k, O_l) = -1 \\ & \text{and } F_{H_2}(O_i, O_j, O_k, O_l) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The function $F_{H_1:H_2}(O_i, O_j, O_k, O_l)$ is nonzero on quadruples for which the two order conjectures make differential predictions. Thus, these latter cross-product expressions are the differences in the number of times the data support the conjecture formalized as H_1 , say A_1 , and the number of times the data support the conjecture formalized as H_2 , say A_2 .

The magnitude of the difference, $A_1 - A_2$, can be evaluated by the same random-relabeling strategy used to assess the adequacy of a single order conjecture. Large positive values of this difference favor H_1 , and large negative values favor H_2 . Normalized indexes would follow the same form as in Equation 5,

$$(A_1 - A_2)/(A_1 + A_2 + A_0), \quad (6)$$

where A_0 refers to the number of "ties" in which H_1 and H_2 make differential order predictions that are unresolved because of tied proximity values in P . Again, this latter normalized measure has zero expectation under random relabeling, is bounded by ± 1 , and can be interpreted as the difference between two conditional probabilities: Among all quadruples (or triples) for which H_1 and H_2 make differential order predictions, the measure in Equation 6 is the probability that we have satisfaction in H_1 and not in H_2 minus the probability that we have satisfaction in H_2 and not in H_1 .

We do not give an explicit numerical application of our strategy for evaluating competing order conjectures in the Holland

context because there appears to be no substantively defensible alternative that is qualitatively different in the literature. At first reading, Gati (1979) seemed to offer a competitive structure by conjecturing that the six personality types are, in effect, best represented by a single partition, $\{(R, I), (A, S), (E, C)\}$; therefore, these 3 pairs should have smaller proximity values than do the other 12, giving a total of 36 order conjectures. With closer inspection, however, Gati's proposal can be reinterpreted as a simple refinement of Figure 1, in which the six objects are not considered equally spaced along the circle; that is, segments RI, AS, and EC are smaller than segments IA, ES, and CR. Thus, our comparison functions, $F_{H_1:H_2}(O_i, O_j, O_k, O_l)$ and $T_{H_1:H_2}(O_i, O_j, O_k)$, will be identically zero because Gati's structure only makes several additional order conjectures that are not explicitly defined within Figure 1. In other words, the three-group partition structure generates no strict order conjecture that is contrary to a strict order conjecture already obtainable from Holland's model.

To evaluate Gati's (1979) proposal apart from Holland's, one could set up either three- or four-argument functions that include only those refinements implicit in the three-group partition, over and above the order conjectures already present in the circular model of Figure 1. In short, unless two qualitatively different structures that lead to diametrically opposite order conjectures for some pairs are hypothesized, there is no clear way of distinguishing which model is "better" by using our particular definition of $F_{H_1:H_2}$, although we might assess whether the refinements implicit in the adoption of one model (e.g., Gati's partition structure) are justifiable from the data by themselves. Although we do not pursue the idea in any detail, the possibility of relaxing the criterion for evaluating two competing conjectures is now based on those order conjectures that are strictly differential. For example, our scoring function might be extended beyond ± 1 and 0 to provide a partial weight of b when $F_{H_1}(O_i, O_j, O_k, O_l) = 1$ and $F_{H_2}(O_i, O_j, O_k, O_l) = 0$ and of $-b$ when $F_{H_1}(O_i, O_j, O_k, O_l) = 0$ and $F_{H_2}(O_i, O_j, O_k, O_l) = 1$ for some b , $0 < b < 1$. Thus, the choice between two competing theories could be made dependent on those cases for which one makes an explicit conjecture and the other does not. Unfortunately, there is an obvious ambiguity in the choice of b and how these partial scoring functions might be justified.

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